Chapter 6a

Basics

Term	Definition/Notes
LAN Overview	private ownership, short distance (≈1km), high speed, low bit error, broadcast transmission, no complex routing, frequently moved machines
LAN Structure	transmission medium, NIC (network interface card), unique MAC "physical" address
LLC (802.2)	Data Link Layer sublayer, protocol multiplexing + error control, 3 service types (Unacknowledged Connectionless is most used)
MAC Protocol	Data Link Layer sublayer, coordinates node medium sharing, reduces/avoids collisions
MAC Address	48-bit globally unique, burned into NIC ROM, identifies LAN nodes

Key Performance Parameters

- $t_{\text{prop}} = d/v$ (propagation delay)
- # wasted bits = $2t_{\text{prop}} * R$ where R is the data transmission bit rate.
- $t_f = L/R$ (frame transmission time, L bits, R bit/s)
- $a = t_{\text{prop}}/t_f$ (normalized delay–bandwidth product, smaller a is better)
- Normalized Delay-Bandwidth Product (unit-less):

$$a = \frac{t_{\text{prop}}}{L/R} = \frac{t_{\text{prop}}}{t_f}$$

- Throughput S: actual successful rate (normalized ≤ 1)
- Max Throughput:

$$R_{\text{eff}} = \frac{L}{t_f + 2t_{\text{prop}}} = \frac{L}{L/R + 2t_{\text{prop}}} = \frac{1}{1+2a}R \quad \text{(bits/second)}$$

- Efficiency ρ: effective transmission / total channel occupancy
- Max Possible Efficiency (unit-less): (not achievable in practice)

$$\rho_{\max} = \frac{L}{L + 2t_{\text{prop}}R} = \frac{1}{1 + 2t_{\text{prop}}R/L} = \frac{1}{1+2a}$$

- Load G: attempts per t_f time unit (core for random access)

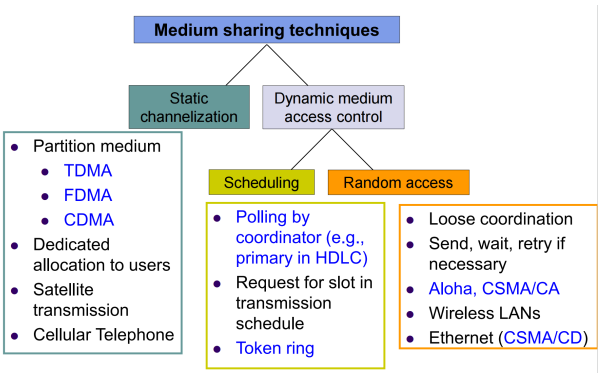

Figure 1: medium sharing

Major Media Sharing Methods

Type	Static	Dynamic	Random
Description	Pre-divide channel	Unified scheduling	Transmit when ready
Pros	No collision, deterministic	High utilization, fairness, QoS	Simple, bursty, low cost
Cons	Resource waste	High complexity, overhead	High collision at load
Apps	Satellite/cellular	Industrial control	Ethernet/WiFi
Examples	TDMA / FDMA / CDMA	Polling / Token / Reservation	ALOHA / CSMA / CSMA-CD / CA

Random Access Core Mechanisms

- *Carrier Sensing*: Detect channel before tx, shortens vulnerable period.
- *Collision Detection (CD)*: Sense while tx, stop + jam if collision.
- *Collision Avoidance (CA)*: Wireless, RTS/CTS + pre-backoff.
- *Backoff*: Random wait; CSMA-CD uses **Binary Exponential Backoff**.

Random Access Protocols

ALOHA Series

Protocol	Vuln. Period	Throughput Formula	Max S	Sync
Pure ALOHA	$2X$	$S = Ge^{-2G}$	0.184 ($G = 0.5$)	No
Slotted ALOHA	X	$S = Ge^{-G}$	0.368 ($G = 1$)	Yes

Slotted ALOHA halves vulnerable period, doubles throughput.

CSMA Series (Vulnerable = t_{prop})

Type	Busy Channel Behavior	Performance
1-Persistent	Continuous sensing, tx when idle	Low delay, high collision at heavy load
Non-Persistent	Random backoff, re-sense	Low collision, high utilization, high delay
p-Persistent	Tx with prob p / wait t_{prop} with $1-p$	Balance delay & utilization (WiFi basis)

CSMA/CD (Ethernet 802.3, wired)

Carrier Sensing → Transmission → Collision Detection → Backoff & Retransmit.
Collision detection time: $2t_{\text{prop}}$ (Ethernet requires $t_f > 2t_{\text{prop}}$, i.e. $a < 0.5$).
Efficiency: $\rho = \dfrac{1}{1+6.44a}$ ($\rho \to 100\%$ when $a \to 0$).
More resources will be wasted if delay-bandwidth product is higher.

CSMA/CA (WiFi 802.11, wireless)

Collision *avoidance* (RTS/CTS + random backoff after idle). Adapts to wireless, lower utilization than CD but no collision.

Dynamic Scheduling Protocols

Token Ring (802.5): Token circulates → capture (busy) → transmit → restore token.
Efficiency: $\rho = \dfrac{1}{1+a'}$ (a' = ring latency (bit) / avg frame length).
No collision, deterministic delay; single point of failure.
Polling/Reservation: Central controller polls / nodes reserve first. Suits QoS (LTE/5G).

MAC Delay Performance

Frame transfer delay: From first bit at source to last bit at destination.

- If no waiting we have minimized delay as $t_f + t_{\text{prop}}$
- Min. normalized: $\dfrac{t_f + t_{\text{prop}}}{t_f} = 1 + a$

Throughput: Actual transfer rate (frames/s or bits/s).

- R bits/s, L bits/frame
- $X = L/R$ s/frame (frame duration)
- λ frames/s (arrival rate)
- Load $\rho = \lambda X$ (unit-less)
- Max throughput: $R/L = 1/X$ frames/s

Logical Link Control Services

- Unacknowledged connectionless service: Unnumbered frame mode of HDLC
- Reliable connection-oriented service: Asynchronous balanced mode of HDLC
- Acknowledged connectionless service: Two additional unnumbered frames added to HDLC

Comparisons

- *Max Throughput*: CSMA/CD > CSMA > Slotted ALOHA > Pure ALOHA
- *Media*: wired (CSMA/CD, Token Ring), wireless (ALOHA, CSMA/CA)
- *Optimize* a: ↑ min frame length L (Ethernet 64B) or ↓ d or ↓ R
- IEEE 802.2: LLC enhances service provided by MAC with (1) Multiplexing of different network protocols; (2) Error control
- *IEEE 802*: 802.3 (Ethernet), 802.5 (Token Ring), 802.11 (WiFi)
- Random access: S ↑ with G at light load; collision surges when G exceeds threshold
- *Binary Exponential Backoff*: backoff range doubles with each collision: $[0,1] \to [0,3] \to [0,7] \ldots$
- *Ethernet key requirement*: $X > 2t_{\text{prop}}$ ensures collision detection before end of frame

Chapter 6b

1. 802.3 vs 802.11

802.3: Wired Ethernet standard.
802.11: Wireless LAN (Wi-Fi) standard.

2. Ethernet Frame Types

Three historic formats; today mostly **Ethernet II**:

- **Ethernet II (DIX)**: Has Type field (e.g., IP). Most common.
- **IEEE 802.2/802.3 (LLC)**: Length field + LLC header; complex.
- **Novell 802.3 raw**: Rare.

Analogy: Ethernet II envelope says "what's inside"; 802.3 envelope says "weight" plus inner label.

3. Repeater, Hub, Bridge, Switch, Router

- **Repeater** (Physical): Amplifies signal; no addressing.
- **Hub** (Physical): Multi-port repeater; broadcasts to all.
- **Bridge** (L2): Learns MAC addresses; 2 ports; filters/forwards.
- **Switch** (L2): Multi-port bridge; core of modern LANs.
- **Router** (L3): Uses IP addresses; connects different networks.

Dimension	Repeater/Hub	Bridge/Switch	Router
Core Mechanism	Signal (L1)	MAC filtering (L2)	IP routing (L3)
Traffic	All segments see traffic	Local isolation	Cross-network
Collision domains	1	2 (each port)	2
Broadcast domains	1	1	2
MAC addresses	No	One mgmt MAC	Two MACs + IPs
MAC header change	No	No	Yes

4. How Bridges Learn Locations

Self-learning:

1. Start: empty MAC table.
2. On receiving frame, record **source MAC + port + timestamp**.
3. Forward/filter: if dest MAC known and port different, forward; if same port, drop; if unknown, flood.

5. Loop Prevention (STP)

Why prevent loops? Broadcast storms, MAC table instability.
Solution: Spanning Tree Protocol (STP).

- Elect one **Root Bridge**.
- Each switch selects **Root Port** (best path to root).
- Each segment selects **Designated Port**.
- Remaining ports are **blocked** (logical break).

6. WLAN Operation

Components: Station (phone/PC) and Access Point (AP).
AP sends **beacons**; station scans, authenticates, associates. Data travels via AP.

7. BSS and ESS

- **BSS**: Basic Service Set — one AP + its stations.
- **BSA**: Basic Service Area (coverage).
- **ESS**: Extended Service Set — multiple BSSs with same SSID, connected via distribution system (wired).

8. CSMA/CA vs CSMA/CD

CSMA/CD (wired): Listen while talking; detect collision; stop; random backoff.
CSMA/CA (wireless): Listen before talk; avoid collision (cannot listen while sending due to strong self-signal).
Why not CD in WLAN? Transmitter deafens its own receiver; hidden terminal problem.

9. Hidden & Exposed Terminals

Hidden: A and B both see AP but not each other → collide at AP.
Solution: RTS/CTS handshake.

- A sends RTS (Request to Send) to AP.
- AP replies CTS (Clear to Send); B hears CTS, defers.

Exposed: C hears A transmitting to B, but C could send to D (different destination); C defers unnecessarily. Partially solved by RTS/CTS.

10. PCF vs DCF

DCF (Distributed Coordination Function): CSMA/CA, contention-based, default.
PCF (Point Coordination Function): AP polls stations, contention-free, optional (rare).

11. RTS and CTS

Optional mechanism to reserve channel and solve hidden terminal.

- **RTS**: small frame from sender: "I need to send".
- **CTS**: from receiver: "OK, everyone be quiet for duration".
- Duration field sets NAV on listening stations.

12. Virtual Carrier Sense & NAV

NAV (Network Allocation Vector): counter in each station; set by duration field in RTS/CTS/data frames. While NAV > 0, station considers channel busy *virtually* — even if physical carrier is idle.

13. Ad Hoc vs Infrastructure

Infrastructure: uses AP; all communication via AP; connects to Internet.
Ad Hoc (IBSS): no AP; stations talk directly; temporary, peer-to-peer (e.g., file transfer).

14. SIFS, PIFS, DIFS

Inter-frame spaces (shorter = higher priority):

- **SIFS**: shortest — for ACK, CTS (immediate response).
- **PIFS**: medium — for PCF polling (AP use).
- **DIFS**: longest — for regular data/RTS in DCF.

15. Quick Reference

802.3 (Ethernet): CSMA/CD, switches, frames.
802.11 (Wi-Fi): CSMA/CA, APs, RTS/CTS, NAV.